

Knowledge, Attitudes, and Practices Regarding Thalassemia

A Cross-Sectional Analysis of Carrier Screening & Cascade Testing

Research Team | August 2026

INTRODUCTION

Thalassemia is a severe autosomal recessive blood disorder. Individuals with Thalassemia Major require lifelong blood transfusions and iron chelation therapy.

Since the condition is genetic, transmission is **entirely preventable** if at-risk carrier couples are identified before marriage.

The World Health Organization (WHO) advocates for **pre-marital partner screening** and **cascade screening** (testing extended family of carriers).

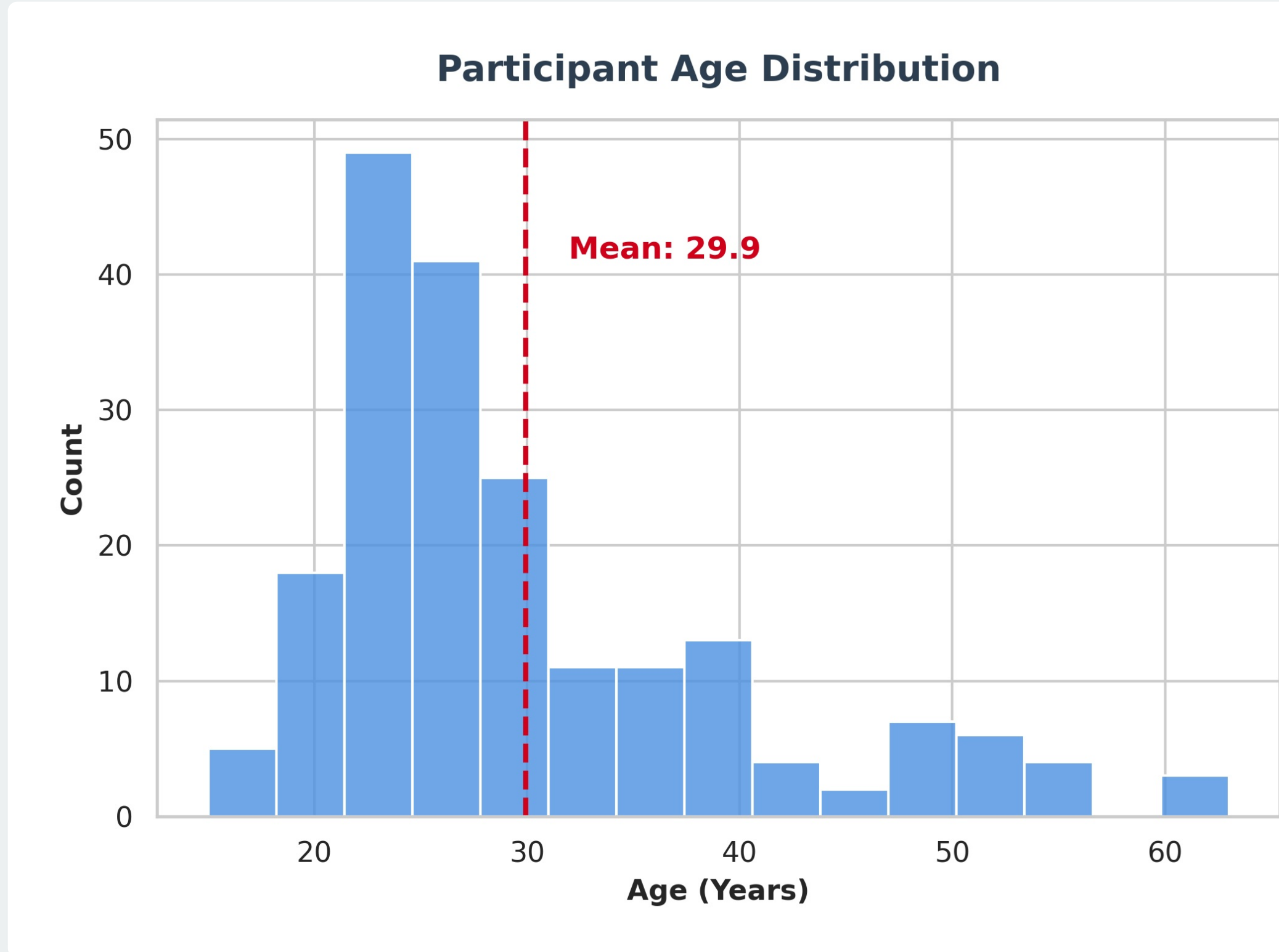
METHODOLOGY

A cross-sectional survey was administered to a localized cohort of **201 participants**. An automated Python/R computational data pipeline was utilized to rigorously process the data.

- Knowledge Score:** A comprehensive 20-point metric evaluating disease transmission and complication awareness.
- Practice Scores:** Partner screening was binarized into 'Safe' (pre-marriage) vs 'Unsafe/Delayed' practices. Cascade testing evaluated relatives' screening frequencies.
- Statistics:** Welch's T-Tests & Pearson's Chi-Square Tests ($\alpha = 0.05$).

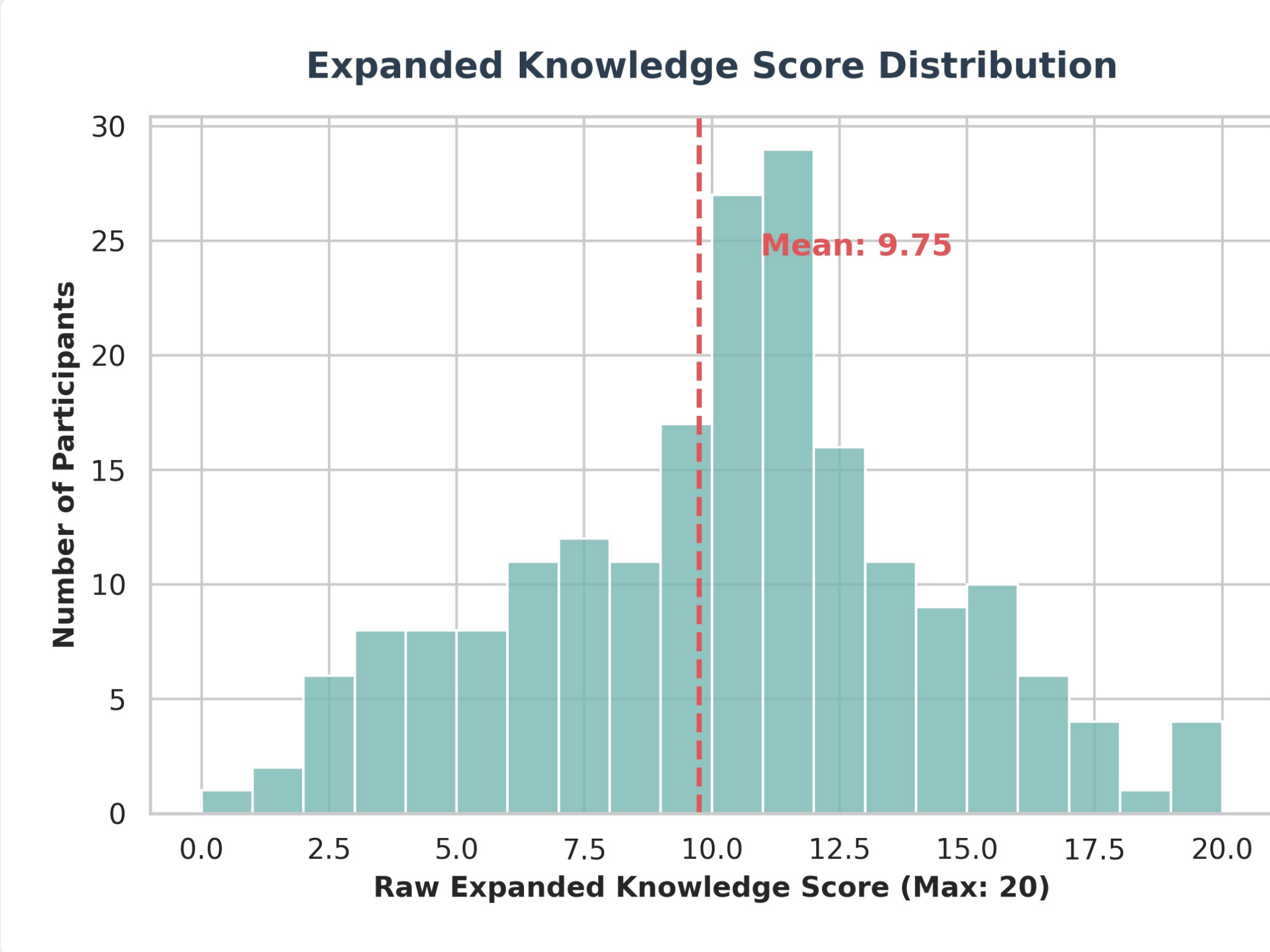
COHORT DEMOGRAPHICS

The study captured a diverse subset of the population, providing robust statistical power.



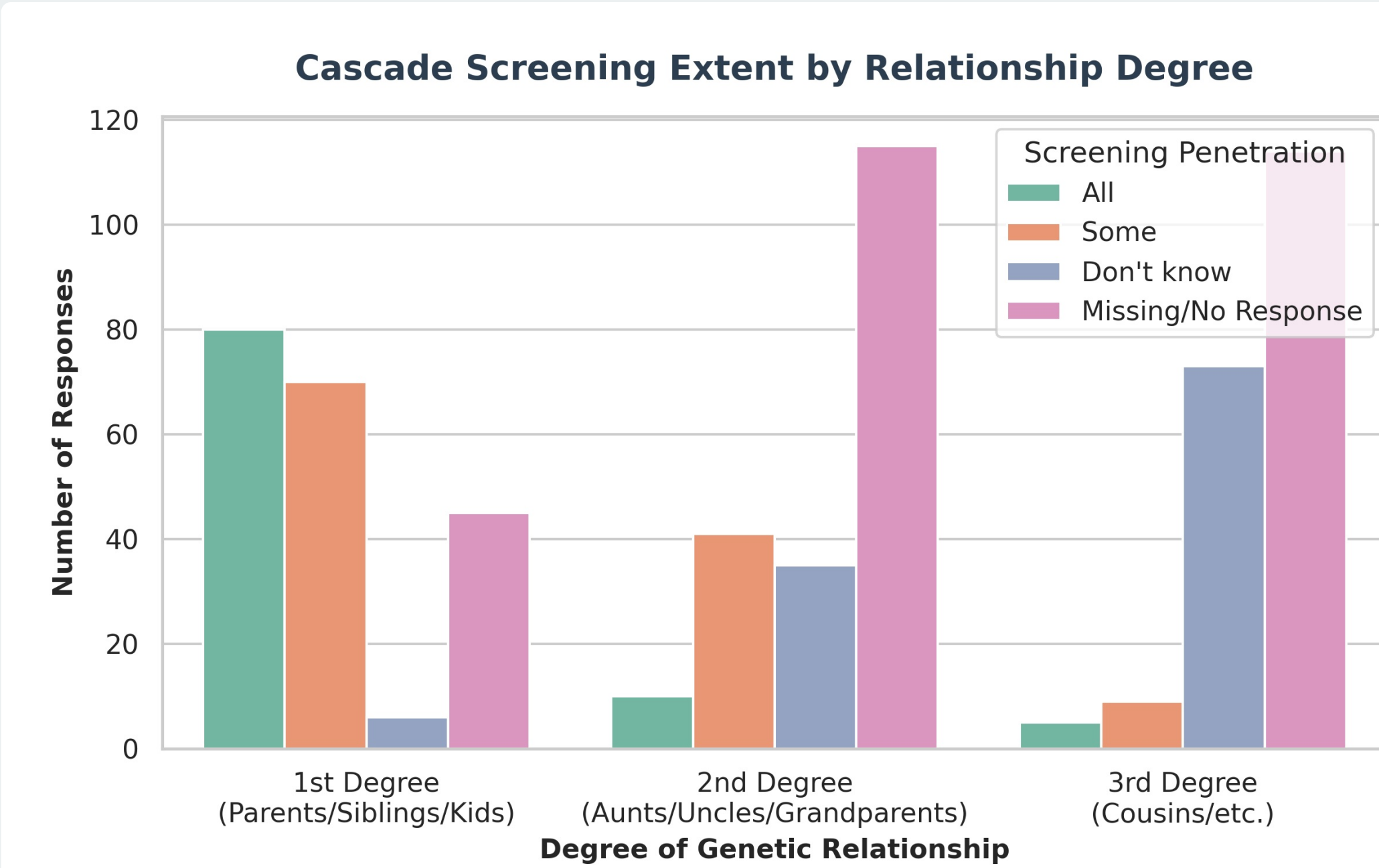
BASELINE KNOWLEDGE

A significant portion of the cohort harbored misconceptions about the disease's curability (via bone marrow transplant) and the 25% transmission risk per pregnancy for carrier couples.



FAMILY SCREENING CASCADE

Screening compliance drops precipitously outside immediate first-degree family, identifying a major gap in disease prevention mechanisms.



KEY INFERENTIAL FINDINGS

Rigorous statistical testing revealed that demographic factors heavily dictate screening outcomes.

Demographic	Impacted Metric	p-value	Significance
Education	Knowledge Score	< 0.001	Degree+ significantly higher
Education	Safe Partner Practice	< 0.001	Degree+ practices safely
Knowledge	Safe Partner Practice	0.019	High Knowledge = Safe Action
Marital Status	Partner Screening	0.004	Marriage dictates timing

Key Insight: Being in the "High Knowledge" category makes an individual statistically more likely to engage in "Safe" (pre-marital) partner screening, proving that education directly modifies behavior.

DISCUSSION & CONCLUSION

- Education is the Catalyst:** Formal education emerged as the single most powerful predictor of safe practices. General awareness campaigns are currently failing lower-education demographics.
- Knowledge Drives Action:** Providing comprehensive disease knowledge successfully bridges the gap into actionable, safe behaviors.
- Strategic Intervention:** To combat transmission effectively, public health frameworks must institutionalize targeted pre-marital counseling, capturing couples during the critical window before marriage.